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GENDER AND SCIENCE*

I. INTRODUCTION

The requirements of . . . correctness in practical judgements and objectivity in theoretical knowledge . . . belong as it were in their form and their claims to humanity in general, but in their actual historical configuration they are masculine throughout. Supposing that we describe these things, viewed as absolute ideas, by the single word 'objective', we then find that in the history of our race the equation objective = masculine is a valid one (George Simmel, quoted by Horney, 1926, p. 200).

In articulating the commonplace, Simmel steps outside of the convention of academic discourse. The historically pervasive association between masculine and objective, more specifically between masculine and scientific, is a topic which academic critics resist taking seriously. Why is that? Is it not odd that an association so familiar and so deeply entrenched is a topic only for informal discourse, literary allusion, and popular criticism? How is it that formal criticism in the philosophy and sociology of science has failed to see here a topic requiring analysis? The virtual silence of at least the non-feminist academic community on this subject suggests that the association of masculinity with scientific thought has the status of a myth which either cannot or should not be examined seriously. It has simultaneously the air of being "self-evident" and "nonsensical" — the former by virtue of existing in the realm of common knowledge (i.e., everyone knows it), and the latter by virtue of lying outside the realm of formal knowledge, indeed conflicting with our image of science as emotionally and sexually neutral. Taken seriously, it would suggest that, were more women to engage in science, a different science might emerge. Such an idea, although sometimes expressed by non-scientists, clashes openly with the formal view of science as being uniquely determined by its own logical and empirical methodology.

The survival of mythlike beliefs in our thinking about science, the very archetype of antimyth, ought, it would seem, to invite our curiosity and demand investigation. Unexamined myths, wherever they survive, have a subterranean potency; they affect our thinking in ways we are not aware of, and to the extent that we lack awareness, our capacity to resist their influence is undermined. The presence of the mythical in science seems

particularly inappropriate. What is it doing there? From where does it come? And how does it influence our conceptions of science, of objectivity, or, for that matter, of gender?

These are the questions I wish to address, but before doing so it is necessary to clarify and elaborate the system of beliefs in which science acquires a gender — which amount to a “genderization” of science. Let me make clear at the outset that the issue which requires discussion is *not*, or at least not simply, the relative absence of women in science. While it is true that most scientists have been, and continue to be, men, the make-up of the scientific population hardly accounts, by itself, for the attribution of masculinity to science as an intellectual domain. Most culturally validated intellectual and creative endeavors have, after all, historically been the domain of men. Few of these endeavors, however, bear so unmistakably the connotation of masculine in the very nature of the activity. To both scientists and their public, scientific thought is male thought, in ways that painting and writing — also performed largely by men — have never been. As Simmel observed, objectivity itself is an ideal which has a long history of identification with masculine. The fact that the scientific population is, even now, a population that is overwhelmingly male, is itself a consequence rather than a cause of the attribution of masculinity to scientific thought.¹ What requires discussion is a *belief* rather than a reality, although the ways in which reality is shaped by our beliefs are manifold, and also need articulating.

How does this belief manifest itself? It used to be commonplace to hear scientists, teachers, and parents assert quite baldly that women cannot, should not, be scientists, that they lack the strength, rigor, and clarity of mind for an occupation that properly belongs to men. Now that the women’s movement has made offensive such naked assertions, open acknowledgment of the continuing belief in the intrinsic masculinity of scientific thought has become less fashionable. It continues, however, to find daily expression in the language and metaphors we use to describe science. When we dub the objective sciences “hard” as opposed to the softer, i.e., more subjective, branches of knowledge, we implicitly invoke a sexual metaphor, in which “hard” is of course masculine and “soft,” feminine. Quite generally, facts are “hard,” feelings “soft.” “Feminization” has become synonymous with sentimentalization. A woman thinking scientifically or objectively is thinking “like a man”; conversely, a man pursuing a nonrational, nonscientific argument is arguing “like a woman.”

The linguistic rooting of this stereotype is not lost among children, who remain perhaps the most outspoken and least self-conscious about

its expression. From strikingly early ages, even in the presence of astereotypic role models, children have learned to identify mathematics and science as male. "Science," my five-year-old son declared, confidently bypassing the fact that his mother was a scientist, "is for men!" The identification between scientific thought and masculinity is so deeply embedded in the culture at large that children have little difficulty internalizing that identification. They grow up not only expecting scientists to be men, but also perceiving scientists as more "masculine" than other male professionals, than, for example, those in the arts. Numerous studies of masculinity and femininity in the professions confirm this observation, with the "harder" sciences as well as the "harder" branches of any profession consistently characterized as more masculine.

In one particularly interesting study of attitudes prevalent among English schoolboys, a somewhat different but critically related dimension of the cultural stereotype emerges. Hudson (1972) observes that scientists are perceived as not only more masculine than are artists, but simultaneously as less sexual. He writes:

The arts are associated with sexual pleasure, the sciences with sexual restraint. The arts man is seen as having a good-looking, well-dressed wife with whom he enjoys a warm sexual relation; the scientists as having a wife who is dowdy and dull, and in whom he has no physical interest. Yet the scientist is seen as masculine, the arts specialist as slightly feminine (p. 83).

In this passage we see the genderization of science linked with another, also widely perceived image of science as antithetical to Eros. These images are not unrelated, and it is important to bear their juxtaposition in mind as we attempt to understand their sources and functions. What is at issue here is the kind of images and metaphor with which science is surrounded. If we can take the use of metaphor seriously, while managing to keep clearly in mind that it is metaphor and language which are being discussed, then we can attempt to understand the influences they might exert — how the use of language and metaphor can become hardened into a kind of reality. One way is through the internalization of these images by scientists themselves, and I will discuss more explicitly how this can happen later in the paper. As a first step, however, the imagery itself needs to be explored further.

If we agree to pursue the implications of attributing gender to the scientific mind, then we might be led to ask, with what or with whom is the sexual metaphor completed? And, further, what is the nature of the act with which

this now desexualized union is consummated? The answer to the first question is immediate. The complement of the scientific mind is, of course, Nature — viewed so ubiquitously as female. “Let us establish a chaste and lawful marriage between Mind and Nature” wrote Bacon (quoted by Leiss, 1972, p. 25), thereby providing the prescription for the birth of the new science. This prescription has endured to the present day — in it are to be found important clues for an understanding of the posture of the virgin groom, of his relation toward his bride, and of the ways in which he defines his mission. The metaphoric marriage of which science is the offspring sets the scientific project squarely in the midst of our unmistakably patriarchal tradition. Small wonder, then, that the goals of science are so persistently described in terms of “conquering” and “mastering” nature. Bacon articulated this more clearly than today’s self-consciousness could perhaps permit when he urged: “I am come in very truth leading you to Nature with all her children to bind her to your service and make her your slave” (Farrington, 1951, p. 197).

Much attention has been given recently to the technological abuses of modern science, and in many of these discussions blame is directed toward the distortions of the scientific program intrinsic in its ambition to dominate nature — without, however, offering an adequate explanation of how that ambition comes to be intrinsic to science. Generally such distortions are attributed to technology, or applied science, which is presumed to be clearly distinguishable from pure science. In the latter, the ambition is supposed to be pure knowledge, uncontaminated by fantasies of control. While it is undoubtedly true that the domination of nature is a more central feature of technology, it is impossible to draw a clear line between pure and applied science. History reveals a most complex relation between the two, as complex perhaps as the interrelation between the dual constitutive motives for knowledge — those of transcendence and power. It would be naïve to suppose that the connotations of masculinity and conquest affect only the uses to which science is put, and leave untouched its very structure.

Science bears the imprint of its genderization not only in the ways it is used, but in the very description of reality it offers — even in the relation of the scientist to that description. To see this, it is necessary to examine more fully the implications of attributing masculinity to the very nature of scientific thought.

Having divided the world into two parts — the knower (mind) and the knowable (nature) — scientific ideology goes on to prescribe a very specific relation between the two. It prescribes the interactions which can consummate

this union, that is, which can lead to knowledge. Not only are mind and nature assigned gender, but in characterizing scientific and objective thought as masculine, the very activity by which the knower can acquire knowledge is also genderized. The relation specified between knower and known is one of distance and separation. It is that between a subject and object radically divided, which is to say no worldly relation. Simply put, nature is objectified. The "chaste and lawful marriage" is consummated through reason rather than feeling, and "observation" rather than "immediate" sensory experience. The modes of intercourse are defined so as to insure emotional and physical inviolability. Concurrent with the division of the world into subject and object is, accordingly, a division of the forms of knowledge into "objective" and "subjective." The scientific mind is set apart from what is to be known, i.e., from nature, and its autonomy is guaranteed (or so it has been traditionally assumed) by setting apart its modes of knowing from those in which that dichotomy is threatened. In this process, the characterization of both the scientific mind and its modes of access to knowledge as masculine is indeed significant. Masculine here connotes, as it so often does, autonomy, separation, and distance. It connotes a radical rejection of any commingling of subject and object, which are, it now appears, quite consistently identified as male and female.

What is the real significance of this system of beliefs, whose structure now reveals a quite intricate admixture of metaphysics, cognitive style, and sexual metaphor? If we reject the position, as I believe we must, that the associations between scientific and masculine are simply "true" — that they reflect a biological difference between male and female brains — then how are we to account for our adherence to them? Whatever intellectual or personality characteristics may be affected by sexual hormones, it has become abundantly clear that our ideas about the differences between the sexes far exceed what can be traced to mere biology; that once formed these ideas take on a life of their own — a life sustained by powerful cultural and psychological forces. Even the brief discussion offered above makes it evident that, in attributing gender to an intellectual posture, in sexualizing a thought process, we inevitably invoke the large world of affect. The task of explaining the associations between masculine and scientific thus becomes, short of reverting to an untenable biological reductionism, the task of understanding the emotional substructure that links our experience of gender with our cognitive experience.

The nature of the problem suggests that, in seeking an explanation of the origins and endurance of this mythology, we look to the processes by which

the capacity for scientific thought develops, and the ways in which those processes are intertwined with emotional and sexual development. By so doing, it becomes possible to acquire deeper insight into the structure and perhaps even the functions of the mythology we seek to elucidate. The route I wish to take proceeds along ground laid by psychoanalysts and cognitive psychologists, along a course shaped by the particular questions I have posed. What emerges is a scenario supported by the insights these workers have attained, and held together, it is to be hoped, by its own logical and intuitive coherence.

II. THE DEVELOPMENT OF OBJECTIVITY

The crucial insight which underlies much of this discussion — an insight for which we are indebted to both Freud and Piaget — is that the capacity for objectivity, for delineating subject from object, is *not* inborn, although the potential for it into doubt is. Rather, the ability to perceive reality “objectively” is acquired as an inextricable part of the long and painful process by which the child’s sense of self is formed. In the deepest sense, it is a function of the child’s capacity for distinguishing self from not-self, “me” from “not-me.” The consolidation of this capacity is perhaps the major achievement of childhood development.

After half a century’s clinical observations of children and adults the developmental picture which emerges is as follows. In the early world of the infant, experiences of thoughts, feelings, events, images, and perceptions are continuous. Boundaries have not yet been drawn to distinguish the child’s internal from external environment; nor has order or structure been imposed on either. The external environment, consisting primarily of the mother during this early period, is experienced as an extension of the child. It is only through the assimilation of cumulative experiences of pleasure and pain, of gratification and disappointment, that the child slowly learns to distinguish between self and other, between image and percept, between subject and object. The growing ability to distinguish his or her self from the environment allows for the recognition of a world of external objects — a world subject to ever finer discrimination and delineation. It permits the recognition of an external reality to which the child can relate — at first magically, and ultimately objectively. In the course of time, the inanimate becomes released from the animate, objects from their perspective, and events from wishes; the child becomes capable of objective thought and perception. The process by which this development occurs proceeds through

sequential and characteristic stages of cognitive growth, stages which have been extensively documented and described by Piaget and his co-workers.

The background of this development is fraught with intense emotional conflict. The primary object which the infant carves out of the matrix of his/her experiences is an emotional object, namely the mother. And along with the emergence of the mother as a separate being comes the child's painful recognition of his/her own separate existence. Anxiety is unleashed, and longing is born. The child (infant) discovers his dependency and need — and a primitive form of love. Out of the demarcation between self and mother arises a longing to undo that differentiation — an urge to re-establish the original unity. At the same time, there is also growing pleasure in autonomy, which itself comes to feel threatened by the lure of an earlier state. The process of emotional delineation proceeds in fits and starts, propelled and inhibited by conflicting impulses, desires, and fears. The parallel process of cognitive delineation must be negotiated against the background of these conflicts. As objects acquire a separate identity, they remain for a long time tied to the self by a network of magical ties. The disentanglement of self from world, and of thoughts from things, requires relinquishing the magical bonds which have kept them connected. It requires giving up the belief in the omnipotence — now of the child, now of the mother — that perpetuates those bonds and learning to tolerate the limits and separateness of both. It requires enduring the loss of a wish-dominated existence in exchange for the rewards of living “in reality.” In doing so, the child moves from the egocentricity of a self-dominated contiguous world to the recognition of a world outside and independent of himself — a world in which objects can take on a “life” of their own.

The recognition of the independent reality of both self and other is a necessary precondition both for science and for love. It may not, however, be sufficient — for either. Certainly the capacity for love, for empathy, for artistic creativity requires more than a simple dichotomy between subject and object. Autonomy too sharply defined, reality too rigidly defined, cannot encompass the emotional and creative experiences which give life its fullest and richest depth. Autonomy must be conceived of more dynamically and reality more flexibly if they are to allow for the ebb and flow of love and play. Emotional growth does not end with the mere acceptance of one's own separateness; perhaps it is fair to say that it begins there. Out of a condition of emotional and cognitive union with the mother, the child gradually gains enough confidence in the enduring reality of both him/herself and the environment to tolerate their separateness and mutual independence. A sense

of self becomes delineated — in opposition, as it were, to the mother. Ultimately, however, both sense of self and of other become sufficiently secure to permit momentary relaxation of the boundary between — without, that is, threatening the loss of either. One has acquired confidence in the enduring survival of both self and other as vitally autonomous. Out of the recognition and acceptance of one's aloneness in the world, it becomes possible to transcend one's isolation, to truly love another.²

The final step — of reintroducing ambiguity into one's relation to the world — is a difficult one. It evokes deep anxieties and fears stemming from old conflicts and older desires. The ground of one's selfhood was not easily won, and experiences which appear to threaten the loss of that ground can be seen as acutely dangerous. Milner (1957), in seeking to understand the essence of what makes a drawing "alive," and conversely, the inhibitions which impede artistic expression, has written with rare perspicacity and eloquence about the dangers and anxieties attendant upon opening ourselves to the creative perception so critical for a successful drawing. But unless we can, the world of art is foreclosed to us. Neither love nor art can survive the exclusion of a dialogue between dream and reality, between inside and outside, between subject and object.

Our understanding of psychic autonomy, and along with it, of emotional maturity, owes a great deal to the work of the English psychoanalyst Winnicott. Of particular importance here is Winnicott's concept of the transitional object — an object intermediate between self and other (as, for example, the baby's blanket). It is called a transitional object insofar as it facilitates the transition from the state of magical union with the mother to autonomy, the transition from belief in omnipotence to an acceptance of the limitations of everyday reality. Gradually, it is given up,

not so much forgotten as relegated to limbo. By this I mean that in health the transitional object does not "go inside" nor does the feeling about it necessarily undergo repression . . . It loses meaning, and this is because the transitional phenomena have become diffused, have become spread out over the whole intermediate territory between "inner psychic reality" and "the external world as perceived by two persons in common," that is to say, over the whole cultural field (Winnicott, 1971, p. 5).

To the diffuse survival of the "creative apperception" he attributes what, "more than anything else, makes the individual feel that life is worth living" (p. 65). Creativity, love, and play are located by Winnicott in the "potential space" between the inner psychic space of "me" and outer social space of "not-me" — "the neutral area of experience which will not be challenged"

(as it was not challenged for the infant) — about which “we will never ask the question: Did you conceive of this or was it presented to you from without” (p. 12).

The inability to tolerate such a potential space leads to psychic distress as surely as the complementary failure to delineate adequately between self and other. “These two groups of people come to us for psychotherapy because in the one case they do not want to spend their lives irrevocably out of touch with the facts of life and in the other because they feel estranged from dream” (p. 67). Both inadequate and excessive delineation between self and other can be seen as defenses, albeit opposite ones, against ongoing anxiety about autonomy.

Emotional maturity, then, implies a sense of reality which is neither cut off from, nor at the mercy of, fantasy; it requires a sufficiently secure sense of autonomy to allow for that vital element of ambiguity at the interface between subject and object. In the words of Loewald (1951), “Perhaps the so-called fully developed, the mature ego is not one that has become fixated at the presumably highest or latest stage of development, having left the others behind it, but is an ego that integrates its reality in such a way that the earlier and deeper levels of ego-reality integration remain alive as dynamic sources of higher organization” (p. 18).

While most of us will recognize the inadequacy of a static conception of autonomy as an emotional ideal, it is easy to fall into the trap of regarding it as an appropriate ideal for cognitive development. That is, cognitive maturity is frequently identified with a posture in which objective reality is perceived and defined as radically divided from the subjective. Our inclination to accept this posture as a model for cognitive maturity is undoubtedly influenced by the definition of objectivity we have inherited from classical science — a definition rooted in the premise that the subject can and should be totally removed from our description of the object. Though that definition has proved unquestionably efficacious in the past, contemporary developments in both philosophy and physics have demonstrated its epistemological inadequacy. They have made it necessary for us to look beyond the classical dichotomy to a more dynamic conception of reality, and a more sophisticated epistemology to support it.

If scientists have exhibited a reluctance to do so, as I think they have, that reluctance should be examined in the light of what we already know about the relation between cognitive and emotional development. Elsewhere (Keller, 1979) I have attempted to show the persistence of demonstrably inappropriate classical ideas even in contemporary physics, where the most

dramatic evidence for the failure of classical ideas has come from. There I try to establish some of the consequences of this persistence, and to account for the tenacity of such ideas. In brief, I argue that the adherence to an outmoded, dichotomous conception of objectivity might be viewed as a defense against anxiety about autonomy of exactly the same kind as we find interfering with the capacity for love and creativity. When even physics reveals "transitional phenomena" — phenomena, that is, about which it cannot be determined whether they belong to the observer or the observed — then it becomes essential to question the adequacy of traditional "realist" modes for cognitive maturity as well as for reality. Our very definition of reality requires constant refinement as we continue in the effort to wean our perceptions from our wishes, our fears, and our anxieties; insofar as our conception of cognitive maturity is dictated by our definition of reality, that conception requires corresponding refinement.

III. THE DEVELOPMENT OF GENDER

What, the reader may ask, has all this to do with gender? Though the discussion has led us on a sizable detour, the implicit argument which relates it to the genderization of science should already be clear. Before articulating the argument explicitly, however, we need an account of the development of gender identity and gender identifications in the context of the developmental picture I have presented thus far.

Perhaps the single most important determinant of our conceptions of male and female is provided by our perceptions of and experiences with our parents. While the developmental processes described above are equally relevant for children of both sexes, their implications for the two sexes are bound to differ. The basic and fundamental fact that it is, for most of us, our mothers who provide the emotional context out of which we forget the discrimination between self and other inevitably leads to a skewing of our perceptions of gender. As long as our earliest and most compelling experiences of merging have their origin in the mother-child relation, it appears to be inevitable that that experience will tend to be identified with "mother," while delineation and separation are experienced as a negation of "mother," as "not-mother." In the extrication of self from other, the mother, beginning as the first and most primitive subject, emerges, by a process of effective and affective negation, as the first object.³ The very processes (both cognitive and emotional) which remind us of that first bond become colored by their association with the woman who is, and forever

remains, the archetypal female. Correspondingly, those of delineation and objectification are colored by their origins in the process of separation *from* mother; they become marked, as it were, as "not-mother." The mother becomes an object, and the child a subject, by a process which becomes itself an expression of opposition to and negation of "mother."

While there is an entire world which exists beyond the mother, in the family constellation with which we are most familiar, it is primarily the father (or the father figure) toward whom the child turns for protection from the fear of re-engulfment, from the anxieties and fears of disintegration of a still very fragile ego. It is the father who comes to stand for individuation and differentiation — for objective reality itself; who indeed can represent the "real" world by virtue of being *in* it.

For Freud, reality becomes personified by the father during the oedipal conflict; it is the father who, as the representative of external reality, harshly intrudes on the child's (i.e., boy's) early romance with the mother — offering his protection and future fraternity as the reward for the child's acceptance of the "reality principle." Since Freud, however, it has become increasingly well understood that the rudiments of both gender and reality are established long before the oedipal period, and that reality becomes personified by the father as soon as the early maternal bond comes to be experienced as threatening engulfment, or loss of ego boundaries. A particularly pertinent discussion of this process is presented by Loewald (1951), who writes:

Against the threatening possibility of remaining in or sinking back into the structureless unity from which the ego emerged, stands the powerful paternal force. . . . While the primary narcissistic identity with the mother forever constitutes the deepest unconscious origin and structural layer of ego and reality, and the motive force for the ego's 'remarkable striving toward unification, synthesis,' — this primary identity is also the source of the deepest dread, which promotes, in identification with the father, the ego's progressive differentiation and structuralization of reality (pp. 15, 17).

Thus it is that, for all of us — male and female alike — our earliest experiences incline us to associate the affective and cognitive posture of objectification with masculine, while all processes which involve a blurring of the boundary between subject and object tend to be associated with the feminine.

The crucial question of course is: What happens to these early associations? While the patterns which give rise to them may be quasi-universal (though strongest, no doubt, in our own form of nuclear family), the conditions which sustain them are not. It is perhaps at this point that specific cultural forces intrude most prominently. In a culture which validates subsequent adult experiences that transcend the subject-object divide, as we find for

example in art, love, and religion, these early identifications can be counteracted — provided, that is, that such experiences are validated as essentially human rather than as “feminine” experience. However, in a culture such as ours, where primary validation is accorded to a science which has been premised on a radical dichotomy between subject and object, and where all other experiences are accorded secondary, “feminine” status, the early identifications can hardly fail to persist. The genderization of science — as an enterprise, as an intellectual domain, as a world view — simultaneously reflects and perpetuates associations made in an earlier, prescientific era. If true, then an adherence to an objectivist epistemology, in which truth is measured by its distance from the subjective, has to be re-examined when it emerges that, by this definition, truth itself has become genderized.

It is important to emphasize, even repeat, that what I have been discussing is a system of beliefs about the meaning of masculine and feminine, rather than any either intrinsic or actual differences between male and female. Children of both sexes learn essentially the same set of ideas about the characteristics of male and female — how they then make use of these ideas in the development of their gender identity as male or female is another question. The relation between the sexual stereotypes we believe in and our actual experience and even observation of gender is a very complex one. It is crucial, however, to make a vigilant effort to distinguish between belief and reality, even, or especially, when the reality which emerges is so influenced by our beliefs. I have not been claiming, for example, that men are by nature more objective, better suited for scientific work, nor that science, even when characterized by an extreme objectivist epistemology, is intrinsically masculine. What I have been discussing are the reasons we might believe all of the above to be true. These beliefs may in fact lead to observed differences between the sexes, though the question of actual differences between men and women in a given culture is ultimately an empirical one. The subsequent issue of how those possible differences might be caused by cultural expectations is yet a separate issue, and requires separate discussion. Without getting into the empirical question of sex differences, about which there is a great deal of debate, it seems reasonable to suggest that we ought to expect that our early beliefs about gender will be subject to some degree of internalization.

To return, then, to the issue of gender development, it is important to recognize that, although children of both sexes must learn equally to distinguish self from other, and have essentially the same need for autonomy, to the extent that boys rest their very sexual identity on an opposition to

what is both experienced and defined as feminine, the development of their gender identity is likely to accentuate the process of separation. As boys, they must undergo a twofold "disidentification from mother" (Greenson, 1968) — first for the establishment of a self-identity, and second for the consolidation of a male gender identity. Further impetus is added to this process by the external cultural pressure on the young boy to establish a stereotypic masculinity, now culturally as well as privately connoting independence and autonomy. The cultural definitions of masculine as what can never appear feminine, and of autonomy as what can never be relaxed, conspire to reinforce the child's earliest associations of female with the pleasures and dangers of merging, and male with both the comfort and the loneliness of separateness. The boy's internal anxiety about both self and gender is here echoed by the cultural anxiety; together they can lead to postures of exaggerated and rigidified autonomy and masculinity which can — indeed which may be designed to — defend against that anxiety and the longing which generates it. Many psychoanalysts have come to believe that, because of the boy's need to switch his identification from the mother to the father, his sense of gender identity tends always to be more fragile than the girl's. Her sense of self-identity may, however, be comparatively more vulnerable. It has been suggested that the girl's development of a sense of separateness may be to some degree hampered by her ongoing identification with her mother. Although she too must disentangle herself from the early experience of oneness, she continues to look toward her mother as a model for her gender identity. Whatever vicissitudes her relation to her mother may suffer during subsequent development, a strong identification based on common gender is likely to persist — her need for "disidentification" is not so radical. Cultural forces may further complicate her development of autonomy by stressing dependency and subjectivity as feminine characteristics. To the extent that such traits become internalized, they can be passed on through the generations by leading to an accentuation of the symbiotic bond between mother and daughter (see, e.g., Chodorow, 1974).

It would seem, then, appropriate to suggest that one possible outcome of these processes is that boys may be more inclined toward excessive and girls toward inadequate delineation — growing into men who have difficulty loving and women who retreat from science. What I am suggesting, then, and indeed trying to describe, is a network of interactions between gender development, a belief system which equates objectivity with masculinity, and a set of cultural values which simultaneously elevates what is defined as scientific and what is defined as masculine. The structure of this network

is such as to perpetuate and exacerbate distortions in *any* of its parts — including the acquisition of gender identity.

IV. THE DEVELOPMENT OF SCIENTISTS

Whatever differences between the sexes such a network might, however, generate — and, as I said earlier, the existence of such differences remains ultimately an empirical question — they are in any case certain to be overshadowed by the inevitably large variations that exist within both the male and female populations. Not all men become scientists, and we must ask whether a science which advertises itself as revealing a reality in which subject and object are unmistakably distinct does not offer special comfort to those who, as individuals (be they male or female), retain particular anxiety about the loss of autonomy. In short, if we can take the argument presented thus far seriously, then we must follow it through yet another step. Would not a characterization of science which appears to gratify particular emotional needs give rise to a self-selection of scientists — a self-selection which would, in turn, lead to a perpetuation of that characterization? Without attempting a detailed discussion of either the appropriateness of the imagery with which science is advertised, or of the personality characteristics which such imagery might select for, it seems reasonable to suggest that such a selection mechanism ought inevitably to operate. The persistence of the characterization of science as masculine, as objectivist, as autonomous of psychological as well as of social and political forces would then be encouraged, through such selection, by the kinds of emotional satisfaction it provides.

If so, the question which then arises is whether, statistically, scientists do indeed tend to be more anxious about their affective as well as cognitive autonomy than nonscientists. Although it is certainly part of the popular image of scientists that they do, the actual measurement of personality differences between scientists and nonscientists has proved to be extremely difficult; it is as difficult, and subject to as much disagreement, as the measurement of personality differences between the sexes. One obvious difficulty arises out of the ambiguity of the term scientist, and the enormous heterogeneity of the scientific population. Apart from the vast differences among individuals, characteristics vary across time, nationality, discipline, and, even, with degree of eminence. The Einsteins of history fail, virtually by definition, to conform to more general patterns either of personality or of intellect. Nevertheless, certain themes, however difficult they may be to pin down, continually re-emerge with enough prominence to warrant

consideration. These are the themes, or stereotypes, on which I have concentrated throughout this paper, and though they can neither exhaustively nor even accurately describe science or scientists as a whole — as stereotypes never can — they do acquire some corroboration from the (admittedly problematic) literature on the “scientific personality.” It seems worth noting, therefore, several features which seem to emerge from a number of efforts to describe the personality characteristics which tend to distinguish scientists from nonscientists.

I have already referred to the fact that scientists, particularly physical scientists, score unusually high on “masculinity” tests, meaning only that, on the average, their responses differ greatly from those of women. At the same time, studies (e.g., Roe, 1953, 1956) report that they tend overwhelmingly to have been loners as children, to be low in social interests and skills, indeed to avoid interpersonal contact. McClelland's subsequent studies confirm these impressions. He writes, “And it is a fact, as Anne Roe reports, that young scientists are typically not very interested in girls, date for the first time late in college, marry the first girl they date, and thereafter appear to show a rather low level of heterosexual drive” (1962, p. 321) (by which he presumably means sexual, thereby confirming, incidentally, the popular image of scientists as “asexual” which I discussed earlier). One of McClelland's particularly interesting findings was that 90% of a group of eminent scientists see, in the “mother-son” picture routinely given as part of the Thematic Apperception Test, “the mother and son going their separate ways” (p. 323) — a relatively infrequent response to this picture in the general population. It conforms, however, with the more general observation (emerging from biographical material) of a distant relation to the mother,⁴ frequently coupled with “open or covert attitudes of derogation” (Roe, 1956, p. 215).

Though these remarks are admittedly sketchy, and by no means constitute a review of the field, they do suggest a personality profile which seems admirably suited to an occupation seen as simultaneously masculine and asexual. Bacon's image of a “chaste and lawful marriage” becomes remarkably apt insofar as it allows the scientist both autonomy and mastery⁵ in his marriage to a bride kept at safe, “objectified” remove.

CONCLUSION

It is impossible to conclude a discussion of the genderization of science without making some brief comments on its social implications. The linking of scientific and objective with masculine brings in its wake a host of secondary

consequences which, however self-evident, may nevertheless need articulating. Not only does our characterization of science thereby become colored by the biases of patriarchy and sexism, but simultaneously our evaluation of masculine and feminine becomes affected by the prestige of science. A circular process of mutual reinforcement is established in which what is called scientific receives extra validation from the cultural preference for what is called masculine, and, conversely, what is called feminine — be it a branch of knowledge, a way of thinking, or woman herself — becomes further devalued by its exclusion from the special social and intellectual value placed on science and the model science provides for all intellectual endeavors. This circularity not only operates on the level of ideology, but is assisted by the ways in which the developmental processes, both for science and for the child, internalize ideological influences. For each, pressures from the other operate, in the ways I have attempted to describe, to create distortions and perpetuate caricatures.

Neither in emphasizing the self-sustaining nature of these beliefs, nor in relating them to early childhood experience, do I wish to suggest that they are inevitable. On the contrary, by examining their dynamics I mean to emphasize the existence of alternative possibilities. The disengagement of our thinking about science from our notions of what is masculine could lead to a freeing of both from some of the rigidities to which they have been bound, with profound ramifications for both. Not only, for example, might science become more accessible to women, but, far more importantly, our very conception of “objective” could be freed from inappropriate constraints. As we begin to understand the ways in which science itself has been influenced by its unconscious mythology, we can begin to perceive the possibilities for a science not bound by such mythology.

How might such a disengagement come about? To the extent that my analysis rests on the crucial importance of the gender of the primary parent, changing patterns of parenting could be of special importance.⁶ But other developments might be of equal importance. Changes in the ethos that sustains our beliefs about science and gender could also come about from the current pressure, largely politically inspired, to re-examine the traditionally assumed neutrality of science, from philosophical exploration of the boundaries or limitations of scientific inquiry, and even, perhaps especially, from events within science itself. Both within and without science, the need to question old dogma has been pressing. Of particular interest among recent developments *within* science is the growing interest among physicists in a process description of reality — a move inspired by, perhaps even necessitated

by, quantum mechanics. In these descriptions object reality acquires a dynamic character, akin to the more fluid concept of autonomy emerging from psychoanalysis. Bohr himself perspicaciously provided us with a considerably happier image than Bacon's — one more apt even for the future of physics — when he chose for his coat of arms the yin-yang symbol, over which reads the inscription: *Contraria Sunt Complementa*.

Where, finally, has this analysis taken us? In attempting to explore the significance of the sexual metaphor in our thinking about science, I have offered an explanation of its origins, its functions, and some of its consequences. Necessarily, many questions remain, and it is perhaps appropriate, by way of concluding, to articulate some of them. I have not, for example, more than touched on the social and political dynamics of the genderization of science. This is a crucial dimension which remains in need of further exploration. It has seemed to me, however, that central aspects of this problem belong in the psychological domain, and further, that this is the domain which tends to be least accounted for in most discussions of scientific thought.

Within the particular model of affective and cognitive development I have invoked, much remains to be understood about the interconnections between cognition and affect. Though I have, throughout, assumed an intimate relation between the two, it is evident that a fuller and more detailed conception is necessary.

Finally, the speculations I offer raise numerous questions of historical and psychological fact. I have already indicated some of the relevant empirical questions in the psychology of personality which bear on my analysis. Other questions of a more historical nature ought also to be mentioned. How, e.g., have conceptions of objectivity changed with time, and to what extent have these conceptions been linked with similar sexual metaphors in other, prescientific eras, or, for that matter, in other, less technological cultures? Clearly, much remains to be investigated; perhaps the present article can serve to provoke others to help pursue these questions.

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NOTES

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¹ For a further elaboration of this theme, see 'Women in Science: A Social Analysis' (Keller, 1974).

² See, e.g., Kernberg (1977) for a psychoanalytic discussion of the prerequisites for mature love.

³ To the extent that she personifies nature, she remains, for the scientific mind, the final object as well.

⁴ These studies are, as is evident, of male scientists. It is noteworthy, however, that studies of the relatively small number of female scientists reveal in women scientists a similar, perhaps even more marked, pattern of distance in relation to the mother. For most, the father proved to be the parent of major emotional and intellectual importance (see, e.g., Plank and Plank, 1954).

⁵ Earlier I pointed out how Bacon's marital imagery constitutes an invitation to the "dominance of nature." A fuller discussion of this posture would also require consideration of the role of aggression in the development of object relations and symbolic thought processes — an aspect which has been omitted from the present discussion. Briefly, it can be said that the act of severing subject from object is experienced by the child as an act of violence, and it carries with it forever, on some level, the feeling tone of aggression. For insight into this process we can turn once again to Winnicott, who observes that "it is the destructive drive that creates the quality of externality" (p. 93), that, in the creation and recognition of the object there is always, and inevitably, an implicit act of destruction. Indeed, he says, "it is the destruction of the object that places the object outside the area of the subject's omnipotent control" (p. 90). Its ultimate survival is, of course, crucial for the child's development. "In other words, because of the survival of the object, the subject may now have started to live a life in the world of objects, and so the subject stands to gain immeasurably; but the price has to be pain in acceptance of the ongoing destruction in unconscious fantasy relative to object-relating" (p. 90). It would seem likely that the aggressive force implicit in this act of objectification must make its subsequent appearance in the relation between the scientist and his object, i.e., between science and nature.

⁶ In this I am joined by Dinnerstein (1976), who has recently written an extraordinarily provocative analysis of the consequences of the fact that it is, and has always been, the mother's "hand that rocks the cradle." Her analysis, though it goes much further and much deeper than the sketch provided here, happily corroborates my own in the places where they overlap. She concludes that the human malaise resulting from the present sexual arrangements can be cured only by dividing the nurturance and care of the infant equally between the mother and the father. Perhaps that is true. I would, however, argue that, at least for the particular consequences I have discussed here, other changes might be of comparable importance.

REFERENCES

- Chodorow, N. (1974), 'Family Structure and Feminine Personality,' in *Woman, Culture and Society*, ed. M. Z. Rosaldo & L. Lamphere (Stanford: Stanford University Press), pp. 43–46.
- Dinnerstein, D. (1976), *The Mermaid and the Minotaur* (New York: Harper & Row).
- Farrington, B. (1951), *Temporis Partus Masculus*, an Untranslated Writing of Francis Bacon. *Centaurus* 1 193–205.

- Greenson, R. (1968), 'Disidentifying from Mother: Its Special Importance for the Boy,' *Explorations in Psychoanalysis* (New York: International Universities Press, 1978), pp. 305–312.
- Horney, K. (1926), 'The Flight from Womanhood,' in *Women and Analysis*, ed. J. Strouse (New York: Dell, 1975), pp. 199–215.
- Hudson, L. (1972), *The Cult of the Fact* (New York: Harper & Row).
- Keller, E. F. (1974), 'Women in Science: A Social Analysis,' *Harvard Magazine*, October, pp. 14–19.
- Keller, E. F. (1979), 'Cognitive Repression in Contemporary Physics,' *Amer. J. Physics* 47, 718–721.
- Keller, E. F. (1980), 'Lewis Carroll: A Study of Mathematical Inhibition,' *J. Amer. Psychoanalytic Assn* 28, 133–160.
- Keller, E. F. (1980), 'Baconian Science: A Hermaphroditic Birth,' *Philosophical Forum* 11, 000–000.
- Keller, E. F. (1979), 'Nature as "Her",' *Proceedings of the Second Sex Conference*, New York University.
- Keller, E. F. (1980), 'Feminist Critique of Science: A Forward or Backward Move?' *Fundamenta Scientiae*, Summer, 1980.
- Keller, E. F. (1983), 'The Mind's Eye' (with C. R. Grontkowski), this volume.
- Kernberg, O. (1977), 'Boundaries and Structure in Love Relations,' *J. Amer. Psychanal. Assn.* 25, 81–114.
- Leiss, W. (1972), *The Domination of Nature* (Boston: Beacon Press, 1974).
- Loewald, H. (1951), 'Ego and Reality,' *Internat. J. Psycho-Anal.* 32, 10–18.
- McClelland, D. C. (1962), 'On the Dynamics of Creative Physical Scientists,' in *The Ecology of Human Intelligence*, ed. L. Hudson (London: Penguin Books), pp. 309–341.
- Milner, M. (1957), *On Not Being Able to Paint* (New York: International Universities Press).
- Plank, E. N., & Plank, R. (1954), 'Emotional Components in Arithmetic Learning as Seen Through Autobiographies,' *The Psychoanalytic Study of the Child* 9 274–293.
- Roe, A. (1953), *The Making of a Scientist* (New York: Dodd, Mead).
- Roe, A. (1956), *The Psychology of Occupations* (New York: Wiley).
- Winnicott, D. W. (1971), *Playing and Reality* (New York: Basic Books).